[INCH-POUND]
GGG-P-840B
March 7, 1994
SUPERSEDING
GGG-P-840A
January 18, 1963

FEDERAL SPECIFICATION

PUNCHES AND DIES, AND PUNCH AND DIE SETS:

METAL CUTTING, HAND; AND PUNCH, CLIP, SHEET METAL

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand operated, punches and dies which are plier-like tools used for punching holes of various sizes in mild sheet steel.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and classes. The punches and dies shall be of the following types and classes, as specified (see 6.2):

Type I - Circular

Class 1	-	Light duty
Class 2	-	Heavy duty

Type II - Clip

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 5110

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Federal Specifications:

PPP-P-40 - Preservation and Packing of Hand Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

Federal Standards:

FED-STD-H28 - Screw-Threads Standards for Federal Services

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI)

ANSI B 46.1 - Surface Texture; Surface Roughness, Waviness & Lay

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

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American Society for Testing and Materials (ASTM):

ASTM A 29/A 29M-91	- Steel Bars, Carbon and Alloy, Hot-Wrought and Cold-Finished General Requirements
ASTM A 322	- Steel Bars, Alloy, Standard Grades
ASTM A 331	- Steel Bars, Alloy, Cold-Finished
ASTM A 576	- Steel Bars, Carbon, Hot-Wrought, Special Quality
ASTM A 681	- Alloy Tool Steels
ASTM E 18	- Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

3. REQUIREMENTS

3.1 Illustrations. Illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of punches and dies and punches and die sets otherwise conforming to this specification.

3.2 Material. The material shall be made from fine-grain, hot-rolled or cold-finished carbon or alloy steel conforming to any of the following standards: ASTM A 29/A 29M, ASTM A 322, ASTM A 331, ASTM A 576, or ASTM A 681. All forgings shall be of uniform quality, free from defects, imperfections, flash, and irregular or sharp projections and edges. Tools shall be capable of withstanding the tests specified in 4.4.

3.3 Hardness.

3.3.1 Male and female dies.

3.3.1.1 Male dies. Male dies shall have a punch hardened from 44 to 54 on the Rockwell C scale (see 4.4.1). The remainder of the male die (the body or shank) shall be hardened from 25 to 62 on the Rockwell C scale.

3.3.1.2 Female dies. Female dies shall have a cutting face hardened from 50 to 60 on the Rockwell C scale (see 4.4.1). The remainder of the female die (the body) shall be hardened from 25 to 62 on the Rockwell C scale.

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3.3.2 Bolts, nuts, screws, and rollers. Bolts, nuts, screws, and rollers used in the assembly of the punch shall be properly hardened and tempered to resist wear and shear.

3.4 Interchangeability of parts. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances in order that such parts may be completely interchangeable or may be adjusted without requiring modification.

3.5 Screw threads. Screw threads on punches and dies shall be smoothly finished having a class 3 or modified class 3 fit and shall be in accordance with FED-STD-H28.

3.5.1 Finish. The surface finish roughness height shall be 32 microinches when tested as specified in accordance with paragraph 4.4.3.

3.6 Identification marking. Punches shall be marked with the manufacturer's name or identifying symbol, and the state or country of manufacture, unless otherwise specified. All marking shall be engraved, etched, molded, or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

3.7 Type I, circular. The type I, circular punch shall consist essentially of an operating lever, intermediate lever or toggle joint arrangement, frame, side linkage, male die, female die, and key. The punch shall be designed to accomplish the following:

a. Punch various size round or irregular shaped holes by hand operation.

b. Accomplish the punching operation with a minimum of effort.

c. Provide for quick changing of the dies.

d. Transmit the punching force through levers and rollers or a lever and toggle joint connected by a series of bolts or screws placed at specific points between the male die and point of pressure on the operating lever.

e. Preclude the possibility of the operating lever being closed beyond the effective punching point, eliminating pinching the hands of the operator.

f. Provide a male die-holder assembly with either a slip-on or a screw-on device for accepting the male dies.

g. Provide proper clearance between the dies to produce a smooth, straight, and true-size hole.

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3.7.1 Operating lever. The operating lever shall be of drop-forged steel, proportionately shaped, of sufficient length and strength capable of transmitting the required force to the punch through the intermediate lever or toggle joint arrangement. The operating lever shall be provided with a comfortable handgrip.

3.7.2 Intermediate lever or toggle joint arrangement. The intermediate lever or toggle joint arrangement shall be of either drop-forged steel or shall be fabricated from high speed tool steel proportionately shaped. A means shall be provided for proper assembly to insure smooth and easy working of the punch.

3.7.3 Frame. The frame shall be of drop-forged steel, proportionately shaped, having a suitable gap opening, depth of throat, and a handle of sufficient length and strength. A means shall be provided in the portion of the frame above the throat for acceptance of the various size male dies and assembly of the side linkage. The portion of the frame below the throat shall be provided with a threaded hole to engage the thread of the female die. The dies, when assembled in the frame above and below the throat, shall be in proper alignment and perpendicular to the throat. The handle shall be provided with a comfortable handgrip.

3.7.4 Side linkage. The side linkage shall be of either drop-forged steel or of cold-rolled steel. The side linkage shall be fastened to the frame and levers in such a manner as to preclude the possibility of misalignment of the dies working in the frame.

3.7.5 Male die. The male die shall be of high speed tool steel and shall have a center point. The body and end of the male die shall be smoothly machined or ground. A means shall be provided on the body of the die to engage the die holder assembly.

3.7.6 Female die. The female die shall be of high speed tool steel. The outer surface of the dies shall be threaded to engage the threaded hole in the frame below the throat opening. The female dies shall be provided with a hole and a slightly larger hole a suitable distance below the cutting edge for clearance of the slugs. The hole in the female dies shall be smoothly finished. The bottom of the dies shall be slotted to accept the key for adjustments.

3.7.7 Key. The key shall be of high speed tool steel, and shall be designed to engage the key slot in the dies for making proper adjustments.

3.7.8 Bolts, nuts, screws, and rollers. Bolts, nuts, screws, and rollers shall be of high speed tool steel and shall be smoothly machined to fit into the assembly of the punch without either binding or excessive play. They shall be properly assembled and adjusted to insure smooth and easy working of the punch. A means shall be provided to preclude the possibility of the nuts becoming loose under normal working conditions.

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3.7.9 Class 1, punch, light duty. The class 1, light duty punch shall be provided with an adjustable depth gauge. A means shall be provided on the side of the frame for location of the gauge and a knurled-head screw to fasten the gauge in place. The class 1 punches shall be similar to figure 1 and shall have a depth of throat of 1-3/4 inches, ± 1/8 inch, and an overall length of 8-3/8 inches, ± 1/8 inch. The class 1 punches shall be capable of punching 9/32 inch round holes in 16 gauge (0.0598 inch) mild sheet steel. The class 1 punch shall be provided with one or a combination of male and female dies, as specified (see 6.2), capable of punching round holes of 1/16, 5/64, 3/32, 7/64, 1/8, 9/64, 5/32, 11/64, 3/16, 13/64, 7/32, 15/64, 1/4, 17/64, and 9/32 inch in diameter.

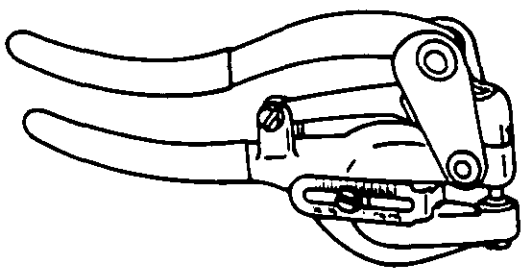


Figure 1. Type I, class 1, circular punch, light duty.

3.7.9.1 Adjustable depth gauge. The adjustable depth gauge shall be of high speed tool steel. The gauge shall have one end bent into the throat and shall be adjustable the length of the throat.

3.7.10 Class 2 punch, heavy duty. The class 2, heavy duty punch shall be similar to figure 2 and shall have a depth of throat of 1-11/16 inches ± 1/8 inch, and a minimum overall length of 23 inches ± 1/2 inch. The mechanical advantage and the method of design of the class 2 punch shall be such that the punch shall be capable of punching 1/4 inch round holes in 1/4 inch mild sheet steel and 1/2 inch round holes in 1/8 inch mild sheet steel. The class 2 punch shall be provided with one or a combination of male and female dies, as specified (see 6.2), capable of punching round holes of 3/32, 7/64, 1/8, 9/64, 5/32, 11/64, 3/16, 13/64, 7/32, 15/64, 1/4, 17/64, 9/32, 19/64, 5/16, 21/64, 11/32, 23/64, 3/8, 25/64, 13/32, 27/64, 7/16, 29/64, 15/32, 31/64, and 1/2 inch in diameter.

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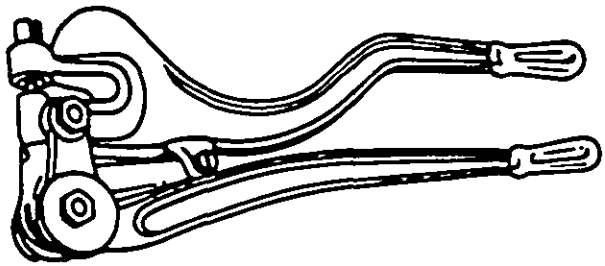


Figure 2. Type I, class 2 circular punch, heavy duty.

3.7.11 Punch sets. Metal punch sets, when required (see 6.2), shall consist of a punch and several male and female dies and shall conform to the requirements of the respective items of classes 1 and 2. Each set shall be furnished in a metal case having provision to accommodate the punch and components with individual holding places for the dies. Cases for punch sets shall be of sheet metal and shall be of durable and rigid construction. The case shall be equipped with a hinged cover and suitable locking device. The case shall be painted or enameled on the inside and outside to prevent corrosion. An instruction sheet containing clear and complete information as to the proper operation of the punch shall be attached to the inside of the case.

3.8 Type II, clip punch. The type II clip punch shall be similar to figure 3 and shall consist essentially of an operating lever, intermediate lever or toggle joint arrangement, frame, side linkage, crimping plunger, plunger lever mechanism and linkage, male die, and female die. The male die and female die, when assembled in the frame, shall be in proper alignment and perpendicular to the throat. The throat of the punch shall be approximately 2 inches in depth. The punch shall have a minimum length of 18 inches and weigh approximately 8 pounds. The punch shall be designed to accomplish the following:

a. To cut, punch, bend, and lock tight an ear approximately 3/16 by 5/16 inch down through three thickness of 20 gauge (0.036 inch) mild sheet steel to fasten the seams together.

b. To accomplish the punching operation smoothly, easily and with a minimum of effort.

c. To permit easy access to cramped places.

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d. To transmit the punching force through levers and rollers or a lever and toggle joint connected by a series of bolts, screws, or pins placed at specific points between the male die and point of pressure on the operating lever.

e. Preclude the possibility of the operating lever being closed beyond the effective punching point, eliminating pinching the hands of the operator.

f. Provide a die holder assembly to accept and securely hold the male and female die.

g. Provide proper clearance between the male and female die to produce a smooth, straight, and true hole.

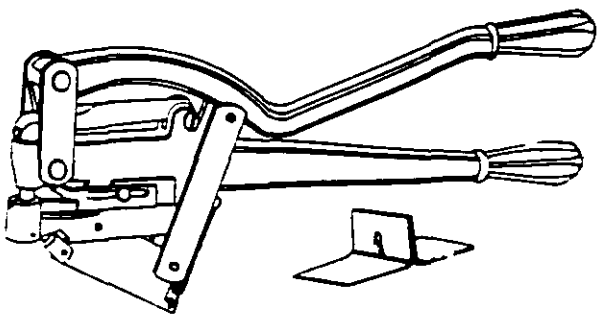


Figure 3. Type II, clip punch.

3.8.1 Operating lever. The operating lever shall conform to the requirements specified in 3.7.1.

3.8.2 Intermediate lever or toggle joint arrangement. The intermediate lever or toggle joint arrangement shall conform to the requirements specified in 3.7.2.

3.8.3 Frame. The frame shall conform to the requirements specified in 3.7.3 except that the portion of the frame below the throat shall have an anvil for the crimping plunger to lock the ear tight and the female die holder shall be properly machined for accepting and holding of the die in such a manner as to insure proper alignment with the male die.

3.8.4 Side linkage. The side linkage shall conform to the requirements specified in 3.7.4.

3.8.5 Crimping plunger. The crimping plunger shall be of sufficient diameter, length, and shall be fastened to the plunger lever in the proper position to push and lock the ear tight.

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3.8.6 Plunger lever mechanism and linkage. The plunger lever mechanism and linkage shall be either drop-forged steel or of cold-rolled steel fastened to the operating lever and frame in such a manner as to crimp and lock the ear tight in the same motion used to cut the ear.

3.8.7 Male die. The male die shall be of high speed tool steel and shall have a center point. The body and punching end of the die shall be smoothly machined or ground. A means shall be provided on the body of the die to engage the dieholder assembly.

3.8.8 Female die. The female die shall be of high speed tool steel. The body of the die shall be machined for acceptance in the frame below the throat opening. The hole in the female die shall be smoothly machined.

3.8.9 Bolts, nuts, screws, or pins. The bolts, nuts, screws, or pins used in the assembly of the punch shall be of high speed tool steel, machined to fit the assembly of the punch. They shall be properly assembled and adjusted to insure easy operation of the punch. A means shall be provided to preclude the possibility of the bolts, nuts, screws, or pins becoming loose under normal working conditions.

3.9 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the specification preparing activity for change to this document.

3.10 Workmanship. Details of workmanship shall be in accordance with the best commercial standards and practices. The item shall be clean and free of corrosion and debris (e.g., chips, shavings, slivers) or other foreign material. The item shall be free from manufacturing workmanship defects (e.g., sharp or rough external edges, corners or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification.

4.2 Sampling procedures. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3 Examination.

4.3.1 Inspection of the end item. Representative samples of punches and dies shall be examined to determine compliance with this specification. A lot shall consist of punches and dies of the same type and class. The sample unit shall be one completely fabricated punch and die.

4.3.1.1 Visual examination of the end item. Visual examination of the end item shall be made for the defects set forth in table I. The Inspection Level shall be I and the Acceptable Quality Level (AQL), expressed in terms of defects per hundred units, shall be 1.5.

TABLE I. End item visual examination defects

Examine	Classification Major	Minor
Type and class (if applicable) of punch not as specified	101	
Tool incomplete; component parts missing or improperly assembled	102	
Materials defective or not as specified; evidence of cracks, deep pits, heavy scale, or flash; evidence of welded or otherwise repaired castings or forgings	103	
Evidence of sticking, binding, or excessive looseness when lever (handle) is operated; bolts, nuts, screws, and pins or rollers not provided with a means to prevent their becoming loose	104	
Punches not smoothly machined and ground, not sharp, not interchangeable (type I only), or cannot be securely attached	105	
Dies defective, punch hole not smoothly finished, cutting edge not sharp, not provided with a larger hole for acceptance of slugs, die not provided with a means for adjustment, or dies not interchangeable; threads stripped, torn, cracked, crossed, drunken, or not perfect full threads (type I only)	106	

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TABLE I. End item visual examination defects (cont'd)

Examine	Classification Major	Minor
Die defective, not securely attached, or cutting edges not sharp and clearly defined (type II only)	107	
Punch and die not in proper alignment or not perpendicular to throat (type I only)	108	
Not provided with a stop to prevent handle movement beyond the effective punching point; operating lever pinches hand	109	
Depth gauge missing, defective, or length of adjustment not as required (type I, class 1 only)	110	
Depth of throat less than the specified minimum (type I only)	111	
Depth of throat exceeds the maximum allowable (type I, class 2 only), or not as required (type II only)		201
Overall length of tool not as required		202
Tool not smoothly finished; evidence of fins, burrs, rust, rough surfaces, or sharp edges		203
Marking, manufacturer's name or trademark missing, illegible, incorrect, or not permanent		204
Set (if applicable) incomplete, carrying case defective, not painted, or not as required (type I, class 1 only)		205
Weight of tool not as required (type II only)		206

4.3.2 Acceptance tests. A sample of punches or dies shall be taken at random from each lot in accordance with MIL-STD-105 at Inspection Level S-4. The Acceptable Quality Level (AQL) is equal to 2.5 percent defective; however, all the sample sizes shall be used and the acceptance number shall be zero until the AQL permits one or greater. Each sample shall be subjected to the applicable tests specified in 4.4 to verify compliance with this specification. Any sample punch or die which does not meet the requirements for any of these characteristics shall be rejected, and if the number of nonconforming punches or dies in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Test procedures.

4.4.1 Hardness test. The dies shall be subjected to a hardness test to determine conformance with 3.3. The Rockwell hardness test shall be conducted in accordance with ASTM E 18.

4.4.2 Punch cutting test.

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4.4.2.1 Type I, circular.

4.4.2.1.1 Class 1, light duty. The class 1, light duty punch shall be tested by punching twenty-five 1/4 inch holes in 16 gauge (0.0598 inch) mild sheet steel.

4.4.2.1.2 Class 2, heavy duty. The class 2 heavy duty punch shall be tested by punching twenty-five 1/4 inch holes in 1/4 inch mild sheet steel.

4.4.2.1.3 Punch cutting test. Two randomly selected sets shall be selected for each lot. A punch cutting test shall be performed on each punch size except for the 1/4 inch size. The test shall be performed on 20 gauge mile steel (nominal 0.035 inch). Three holes shall be punched for each punch size from each set. The acceptance criteria shall be as specified in 4.4.2.3.

4.4.2.2 Type II, clip . The Type II, clip punch shall be tested by cutting, punching, and locking twenty-five ears down through three thickness of 20 gauge (0.036 inch) mild sheet steel.

4.4.2.3 Punch failure. Failure of the punch shall occur when the dies produce a ragged or burred hole, fail to lock the ear, break, chip, or spall. Any failure shall be sufficient reason for rejection of the lot. Rejected lots shall be 100 percent re-inspected for the defective characteristics before being resubmitted to the Government for reinspection.

4.4.3 Surface roughness. Surface roughness specified in paragraph 3.5.1, shall be tested in accordance with ANSI B46.1

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification. The lot size shall be the number of shipping containers in the end item inspection lot. The sample unit shall be one container. The inspection level shall be S-2 and the AQL expressed in terms of defects per hundred units, shall be 4.0.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling Distortion of container

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A or C, as specified (see 6.1).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use.

6.1.1 Type I, circular, class 1 and 2 punches. Type I, circular, class 1 and 2 punches are primarily for use by metalworkers to punch various size holes in mild sheet steel.

6.1.2 Type II, clip punches. Type II, clip punches are primarily for use by metalworkers to cut, punch, bend, and lock tight an ear down through sheet steel to fasten seams together.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type and class required (see 1.2.1).
c. Size of dies required (see 3.7.9 and 3.7.10).
d. Whether punches are required in sets (see 3.7.11).
e. Selection of applicable levels of preservation and packaging, and packing (see 5.1).

MILITARY INTERESTS:

Custodians

Army - GL
Air Force - 99

Review Activity

Air Force - 84

MILITARY COORDINATING ACTIVITY:

Army - GL

Preparing Activity

GSA-FSS

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